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A hybrid RRT-DWA path planning framework for UAVs in dynamic environments

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To address the limitations of single-path planning algorithms in dynamic and complex environments, this paper proposes a hybrid planning framework that integrates global and local planning using a combination of Rapidly-exploring Random Tree (RRT) and Dynamic Window Approach (DWA) algorithms. The global planning employs the RRT algorithm to generate a global path, ensuring goal-directed and near-optimal paths through a cost-grid-based search mechanism. The local planning utilizes an improved DWA algorithm, which, under the guidance of the global path, dynamically adjusts the direction and speed of the UAV in real time to avoid unknown obstacles and adapt to dynamic environmental disturbances. The proposed method addresses key limitations of conventional DWA, including local optima and limited adaptability in dynamic environments. The two planning methods are integrated through a strategy of selecting key sub-goal points: when the UAV follows the global path and encounters a local minimum, the sub-goal points guide the UAV to escape from the local minimum and optimize the path quality. Extensive simulation experiments validate that the hybrid framework significantly enhances the UAV's obstacle avoidance capabilities and task execution efficiency.The address of the ROS-Gazebo experimental project - https://app.theconstruct.ai/rosjects/980801/.The address of MATLAB source code - https://github.com/moxing15/rrt-dwa.

Keywords UAV, Path Planning, Dynamic Window Approach (DWA), Rapidly-exploring Random Tree (RRT)

As UAV technology continues to advance at a rapid pace, its applications are expanding rapidly across military, civilian, and commercial sectors. UAVs are now utilized for patrol[1], surveillance[2], logistics delivery[3], and agricultural plant protection[4]. Thanks to their flexibility, high efficiency, and cost-effectiveness, UAVs are increasingly recognized as a vital component of modern technological solutions. However, the widespread adoption of UAVs also presents significant challenges, with path planning emerging as one of the most critical issues.

Path planning involves UAVs making informed decisions and autonomously devising feasible routes based on mission objectives and environmental data. The development of more efficient and intelligent path planning algorithms for UAVs has become a focal point and a formidable challenge in the field of path planning. The goal is to meet the diverse planning requirements of UAVs in various environments and enhance the safety and reliability of their autonomous operations. UAV path planning algorithms can generally be categorized into two main types: global planning algorithms[5,6] and local planning algorithms[7,8].

Typical global path planning algorithms include the A* algorithm[9], Dijkstra algorithm[10], Probabilistic Roadmap (PRM) algorithm[11], and Rapidly-exploring Random Tree (RRT) algorithm[12]. Graph search-based methods such as the A* algorithm and Dijkstra algorithm are easy to understand and implement, and they have the advantage of fast computation speed. However, they may encounter low computational efficiency in complex environments[13,14]. The PRM algorithm can plan initial paths for mobile robot path planning in limited environments, but it may be accompanied by high spatial and temporal computational complexity when calculating paths, and it may produce unsafe path calculations in complex environments[15]. The RRT algorithm, which breaks through the dimensionality limitations through probabilistic sampling, has shown significant advantages in complex obstacle environments. However, its random expansion mechanism is prone to generating redundant nodes and relies on binary environment representation, ignoring the differences in obstacle threats[16].

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While global planners are effective for generating an overall path in a known environment, they typically lack the reactivity to handle unexpected dynamic obstacles. This limitation necessitates the use of local planners, which focus on real-time obstacle avoidance and trajectory refinement. Unlike the aforementioned global path planning methods, local path planning algorithms focus more on real-time obstacle avoidance and dynamic response capabilities. Common local path planning algorithms include the Artificial Potential Field (APF)[17] and Dynamic Window Approach (DWA)[18]. The APF algorithm has the advantages of clear logic, low computational complexity, and strong global planning ability, but it is prone to falling into local optima and is very sensitive to parameters[19]. As a classic local path planning algorithm, DWA is widely used in dynamic obstacle avoidance tasks for mobile robots, UAVs, and other fields due to its computational efficiency and strong real-time performance[20]. However, the traditional DWA algorithm has inherent short-comings such as the lack of a global path, insufficient adaptability to dynamic environments, and rigid evaluation functions, which lead to problems such as local optimality, path oscillation, or failure of obstacle avoidance in complex scenarios[21].

To address these issues, researchers have proposed various solutions. For example, Kobayashi et al.[22] proposed an improved DWA algorithm combined with Virtual Manipulators. By introducing virtual force fields and dynamic obstacle motion prediction, it enhances the obstacle avoidance capability and path smoothness of traditional DWA in dynamic environments. Dobrevski et al.[23] proposed Dynamic Adaptive DWA, which introduces dynamically adjusted evaluation weights and real-time environmental adaptability strategies to solve the problems of fixed velocity space sampling and rigid evaluation functions that cause path oscillation or stagnation. Hu et al.[24] improved the DWA algorithm by combining pedestrian flow prediction models with multi-robot coordination strategies. It dynamically adjusts the evaluation function weights and velocity sampling space using real-time environmental perception data, solving the problems of low obstacle avoidance efficiency and path oscillation in dense dynamic crowds for traditional DWA. Li et al.[25] used the global path provided by A* as the reference trajectory for DWA, constraining the local velocity sampling space to avoid falling into local optima. Jiang et al.[26] integrated an improved A* algorithm with DWA, using the global path generated by A* to constrain the local velocity sampling space of DWA and introducing a dynamic obstacle prediction mechanism to solve the problems of local optimality and delayed obstacle avoidance response of traditional DWA in dynamic environments.

Although these sensor-enhanced and rule-optimized improvement methods have significantly enhanced the local performance of DWA, its theoretical limitations in adapting to dynamic environments, long-term prediction, and cross-scene generalization are still difficult to break through. This situation has driven a shift in research approaches – with the successful validation of intelligent algorithms such as Genetic Algorithm (GA)[27], Grey Wolf Optimization (GWO)[28], Reinforcement Learning (RL)[29], and Deep Reinforcement Learning (DRL)[30] in UAV path planning, researchers have begun to construct agent-driven new architectures for DWA. For example, Kim et al.[31] proposed an improved DWA algorithm based on Reinforcement Learning (RL). By training agents to dynamically optimize the evaluation function weights and velocity sampling strategies, it enables traditional DWA to adapt to the complex changes in dynamic environments, solving the problems of rigid obstacle avoidance decision-making and poor environmental adaptability under fixed rules. Li et al.[32] proposed a hybrid path planning method that integrates an improved Genetic Algorithm (GA) with an improved DWA. The GA optimizes the global coarse path and generates key waypoints, which are then used by the improved DWA to achieve local re-al-time obstacle avoidance, addressing the issues of insufficient globality in traditional DWA and poor real-time performance of GA. Yang et al.[33] proposed a novel adaptive DWA combined with a Fuzzy Neural Network (FNN). The FNN dynamically learns environmental features and adjusts the evaluation function weights and parameters of DWA in real time, significantly enhancing the adaptive obstacle avoidance capability and decision-making intelligence of traditional DWA in complex dynamic environments. Wu et al.[34] leveraged the global optimization capability of Deep Reinforcement Learning (DRL) and the fast local planning capability of the DWA. By defining the action space of the proximal policy optimization algorithm to match the linear and angular velocity descriptions in DWA, it achieves more optimal collision avoidance decisions.

The integration of intelligent algorithms with DWA has enhanced the adaptability and generalization capabilities of the algorithms. However, due to the data dependency of intelligent algorithms, delays in perception and decision-making, high computational complexity, and limitations of hardware conditions, their application in UAVs often results in limited adaptability to dynamic environments, insufficient real-time obstacle avoidance capabilities, and high computational requirements.

In order to overcome the shortcomings of DWA, such as local minima and insufficient long-term prediction capabilities, improve obstacle avoidance in dynamic environments, and avoid the increased computational complexity brought by the introduction of intelligent algorithms, this paper optimizes DWA based on the dynamic characteristics of UAVs. The evaluation function of DWA is improved, and the planning accuracy and safety are enhanced by integrating global path information and introducing dynamic prediction. RRT-DWA hybrid path planning algorithm is proposed, which combines the global planning of RRT with the local obstacle avoidance capability of DWA to significantly improve the planning ability of UAVs in dynamic and complex environments.

Methods

Analysis and improvement of traditional DWA

Velocity window generation

(1) Velocity Boundary Constraints The velocity boundaries typically include four aspects: minimum linear velocity v_{min}, maximum linear velocity v_{max}, minimum angular velocity ω_{min}, and maximum angular velocity ω_{max}. The velocity constraints are shown in Eq. (1)[18].

$$V_{self} = \{(v, \omega) \mid v_{min} \leq v \leq v_{max}, \omega_{min} \leq \omega \leq \omega_{max}\} \tag{1}$$

where V_{self} represents the velocity window of the system, which includes all possible velocity pairs (v, ω), and v, ω are velocity and angular velocity of the UAV.

(2) Acceleration constraints Assuming the current velocity is v and the angular velocity is ω, within a time step Δt, the mathematical representation of the acceleration constraints is shown in Eq. (2)[18].

$$V_{acc} = \{(\nu, \omega) \mid |\nu - \nu_c| \leq \alpha_\nu \Delta t, |\omega - \omega_c| \leq \alpha_\omega \Delta t\} \tag{2}$$

where V_{acc} represents the velocity window under the acceleration constraints.

(3) Environmental Threat Constraints Environmental threats also limit the motion speed of the UAV, especially when it is close to obstacles. The UAV must reduce its speed to avoid collisions. Equation (3) represents the velocity constraints based on obstacles[18].

$$V_{\text{safe}} = \{(\nu, \boldsymbol{\omega}) \mid \nu < \sqrt{2\text{dist}(\nu, \boldsymbol{\omega}) \cdot \boldsymbol{\alpha_\nu}}, \boldsymbol{\omega} < \sqrt{2\text{dist}(\nu, \boldsymbol{\omega}) \cdot \boldsymbol{\alpha_\omega}}\} \tag{3}$$

where $\text{dist}(v, \omega)$ is the minimum distance between the UAV and the nearest obstacle along the predicted trajectory.

(4) The Formation of The Comprehensive Velocity Window The comprehensive velocity window of the UAV is the intersection of the three constraints: self-velocity boundary constraints, acceleration constraints, and environmental threat constraints, as shown in Eq. (4)[18].

$$V_{\text{window}} = V_{\text{self}} \cap V_{\text{acc}} \cap V_{\text{safe}} \tag{4}$$

Velocity window optimization

(1) Velocity Boundary Constraints For UAVs, the minimum linear velocity v_{min} in Eq. (1) is generally set to 0, indicating that the UAV is stationary. The maximum linear velocity v_{max} is limited by the motor. The minimum angular velocity ω_{min} is usually set to the negative of the maximum angular velocity, while the maximum angular velocity ω_{max} represents the maximum turning speed that a UAV can achieve. In Eq. (2), the range of velocity and acceleration changes is restricted by the maximum linear acceleration $a_{v,max}$ and the maximum angular acceleration $a_{\omega,max}$. The mathematical expression for the acceleration constraints is shown in Eq. (5).

$$V_{acc}^{'} = \{(\nu, \boldsymbol{\omega}) || \alpha_\nu | \leq \alpha_{\nu,max}, |\alpha_\omega| \leq \alpha_{\omega,max}\} \tag{5}$$

(2) Environmental Threat Constraints When the distance between the UAV and the obstacle is too small, the UAV must reduce its speed to avoid a collision. The environmental threat constraints V'_{safe} of the UAV can be mathematically expressed using Eq. (6).

$$V'_{safe} = \{(\nu, \boldsymbol{\omega}) \mid \nu < \sqrt{2(dist(\nu, \boldsymbol{\omega}) + d_{add}) \cdot a_{\nu,max}}, \boldsymbol{\omega} < \sqrt{2(dist(\nu, \boldsymbol{\omega}) + d_{add}) \cdot a_{\omega,max}}\} \tag{6}$$

where $dist(v, \omega)$ is the closest obstacle distance along the predicted trajectory, and d_{add} is the threat distance weight.

Combining Eqs. (1), (4), (5) and (6), the optimized velocity window for the UAV is shown in Eq. (7). The comprehensive velocity window ensures that the UAV can select a safe and effective velocity vector for movement, as illustrated in Fig. 1.

$$V_{win} = V_{self} \cap V'_{acc} \cap V'_{safe} \tag{7}$$

Evaluation function calculation

(1) Goal distance Evaluation Function The goal distance evaluation function is used to measure the distance from the UAV's current position to the goal point under the current velocity. The larger the value, the closer the UAV is to the target. And its mathematical expression is shown in Eq. (8).

$$C_{\text{goal}} = \frac{1}{1 + max(0, d_{\text{goal}} - d_{\text{goal}}^{\max})} \tag{8}$$

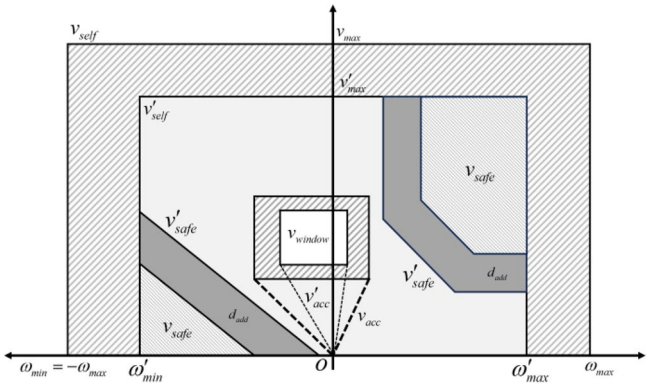


Fig. 1. Improved speed window.

where d_{goal} represents the direct geometric distance from the UAV to its target. $d_{\text{goal}}^{\max}$ is the maximum tolerance radius of the goal point area. C_{goal} is the target distance cost of UAV.

(2) Obstacle Avoidance Evaluation Function The minimum distance $d_{\min}(t)$ between the UAV's trajectory and the obstacle at a certain moment, the minimum distance to the obstacle is represented as shown in Eq. (9).

$$C_{\text{obs}} = \sum_{t} d_{\min}(t) \tag{9}$$

where C_{obs} represents the obstacle avoidance cost of the trajectory at time *t*.

(3) Velocity Smoothness Evaluation Function Considering that abrupt accelerations or decelerations consume excessive energy and increase flight instability, the velocity smoothness cost is represented by the rate of change of velocity at each point along the trajectory. Let the velocity difference between trajectory points be Δv, then the velocity smoothness cost C_{smooth} can be expressed as shown in Eq. (10), indicating that the smaller the velocity change, the better the trajectory.

$$C_{smooth} = \frac{1}{\Delta v} \tag{10}$$

Improvement and optimization of the evaluation function
To adapt to the complex motion characteristics of UAVs and the rapid changes in dynamic environments, this section improves and optimizes the evaluation function. In the improved DWA algorithm, the introduction of dynamic weighting coefficients enhances the performance of the algorithm.

(1) Goal distance Evaluation Function Fixed weights in the evaluation function can lead to myopic behavior. For instance, a high constant goal weight might drive the UAV into a crowded area, while a high constant obstacle weight might cause it to stagnate far from obstacles. To address this, we introduce a dynamic weight $w_{goal}(t)$ that decreases as the UAV approaches the goal point. This allows the UAV to prioritize obstacle avoidance when navigating through cluttered spaces and shift focus to goal point attainment when the path is clear, thereby achieving a better balance between global mission completion and local safety. The dynamic weighting coefficient of the distance to the goal point $w_{goal}(t)$ is dynamically adjusted according to the progress of the task, as shown in Eq. (11).

$$w_{goal}(t) = w_{goal} \cdot e^{-\sigma d_{goal}} \tag{11}$$

where w_{goal} is the initial distance of the goal weight, σ controls the decay rate and d_{goal} represents the straight line distance between the current position and the goal point of the UAV. As the UAV progressively approaches the goal point, $w_{goal}(t)$ tends to 0, indicating that the cost of the distance to the goal point becomes less important in the decision-making process. Taking into account the dynamic weighting coefficient, the target distance evaluation function is shown in Eq. (12).

$$C_{goal} = w_{goal}(t) \cdot \frac{1}{d_{goal} - d_{goal}^{max}} \tag{12}$$

(2) Obstacle Avoidance Evaluation Function The dynamic weighting coefficient $w_{obs}(t)$ for obstacle avoidance is used to adjust the weight of the obstacle avoidance cost. It is dynamically calculated as shown in Eq. (13).

$$w_{obs}(t) = w_{obs} \cdot \frac{d_{min}(t)}{d_{min}(t) + \gamma} \tag{13}$$

where $d_{min}(t)$ is the current minimum separation between the UAV and the closest obstacle, w_{obs} is the initial obstacle avoidance weight and γ controls the decay rate. Then the obstacle avoidance evaluation function is shown in Eq. (14).

$$C_{obs} = w_{obs}(t) \cdot \sum_{i=1}^{n} d_{min}(t) \tag{14}$$

where d_{min} is the minimum distance from the $i-th$ obstacle to the predicted trajectory of UAV, $\lambda_{dynamic}$ is a coefficient that scales the influence of dynamic obstacles, typically set to a value greater than 1 to prioritize avoiding moving threats.

(3) Velocity Smoothness Evaluation Function The dynamic weighting coefficient for velocity smoothness $w_{smooth}(t)$ reflects the impact of the velocity smoothness cost on the planning process. It is defined as shown in Eq. (15).

$$w_{smooth}(t) = w_{smooth} \cdot \frac{\kappa}{\kappa + \Delta v} \tag{15}$$

where Δv is the velocity change between adjacent moments, w_{smooth} is the initial smoothness weight, and κ controls the decay rate.

Then the velocity smoothness evaluation function is shown in Eq. (16).

$$C_{smooth} = w_{smooth}(t) \cdot \left(\frac{\Delta v}{v_{max}} + \frac{\Delta a}{a_{max}} \right) \tag{16}$$

where, Δv and $\Delta \alpha$ represent the differences in velocity and acceleration, respectively, between consecutive trajectory points. This term penalizes abrupt changes, promoting smoother trajectories.

Combining Eqs. (12), (14), and (16), the improved evaluation function is shown in Eq. (17).

$$C_{total} = C_{goal} + C_{obs} + C_{smooth} \tag{17}$$

Design of the DWA dynamic path planning algorithm

In practical applications, UAVs often encounter complex flight environments. In multi-task handling, UAVs need to deal with dynamic obstacles and avoid interference with each other. To address this challenge, a dynamic obstacle avoidance mechanism is introduced in the local planning stage. This mechanism takes into account the status and position of other UAV or dynamic obstacles, predicts collision risks in real time, and adjusts the speed and heading of each UAV to ensure the safe completion of the mission.

Velocity smoothness evaluation function

Assuming the motion state of a UAV *i* at the current moment is $x_i(t)$, $v_i(t)$ and $a_i(t)$, the motion state of another UAV (or dynamic obstacle) *j* is $x_j(t)$, $v_j(t)$ and $a_j(t)$. Their positions $x_i(t+T)$ and $x_j(t+T)$ at time $t+T$ can be predicted by Eq. (18).

$$\begin{cases} x_i(t+T) = x_i(t) + v_i(t) \cdot T + \frac{1}{2} a_i(t) \cdot T^2 \\ x_j(t+T) = x_j(t) + v_j(t) \cdot T + \frac{1}{2} a_j(t) \cdot T^2 \end{cases} \tag{18}$$

Calculate the relative position $r_{ij}(t+T)$ of *i* and *j* at time $t+T$ by Eq. (19).

$$r_{ij}(t+T) = x_i(t+T) - x_j(t+T) \tag{19}$$

Predict the relative position $r_{ij}(t+T)$ and calculate its relationship with the safety distance d_{safe}. When Eq. (20) is satisfied.

$$\|r_{ij}(t+T)\| < d_{safe} \tag{20}$$

A collision is likely to occur, triggering the obstacle avoidance mechanism.

Real-time obstacle avoidance strategy
When an imminent collision is detected, the UAV's velocity is adjusted to increase the distance from the dynamic obstacle. Assuming the minimum distance between UAV *i* and the obstacle (or another UAV) at the future time $t+T$ is $r_{ij}(t+T)$, the risk of collision can be reduced by decelerating. The deceleration strategy is shown in Eq. (21).

$$v_i(t+T) = v_i(t) - \Delta v_i \tag{21}$$

where Δv_i is the amount of velocity reduction, and the magnitude of deceleration is shown in Eq. (22).

$$\Delta v_i = \left| \frac{d_{safe} - r_{ij}(t+T)}{r} \right| \tag{22}$$

This formulation demonstrates that adequately decreasing the UAV's velocity can restore the minimum safety distance to within acceptable thresholds. The adjustment direction is typically determined by the obstacle avoidance vector, which can be derived by computing the normal vector $\hat{r}_{\hat{y}}(t+T)$ of the relative position between the UAV and obstacles, as formulated in Eq. (23).

$$\hat{r}_{\hat{y}}(\iota+T) = \frac{r_{\hat{y}}(\iota+T)}{\|r_{\hat{y}}(\iota+T)\|} \tag{23}$$

Based on the predetermined obstacle avoidance vector, the UAV computes its required turning angle to ensure trajectory divergence from obstacles. The corresponding angular velocity for heading adjustment is then derived as specified in Eq. (24).

$$\omega_i(\iota) = \frac{\hat{r}_{\hat{y}}(\iota+T) \cdot \nabla\theta}{T} \tag{24}$$

where $\nabla\theta$ is the direction of the adjustment angle, and $\omega_i(t)$ is the angular velocity of the UAV.

Design of the RRT-DWA hybrid path planning algorithm

The RRT-DWA hybrid path planning algorithm fully utilizes the advantages of global planning and local obstacle avoidance, ensuring the rationality of the global path while enhancing the ability to handle dynamic obstacles.

Sub-goal point selection and local minimum optimization
Once the global RRT path planning is generated, selecting appropriate sub-goal points can improve the efficiency of local planning. To ensure that the selected sub-goal points are representative, they can be determined through spatial geometric relationship analysis. Assuming $P_i(x_i, y_i)$ and $P_{i+1}(x_{i+1}, y_{i+1})$ are adjacent path points, the Euclidean distance *d* between path points is shown in Eq. (25).

$$d = \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \tag{25}$$

The angle change θ between adjacent points on the path can be calculated using Eq. (26).

$$\theta = arctan\left(\frac{y_{i+1} - y_i}{x_{i+1} - x_i}\right) - arctan\left(\frac{y_i - y_{i-1}}{x_i - x_{i-1}}\right) \tag{26}$$

If the distance *d* between path points is greater than a certain set threshold and the angular change θ exceeds a certain critical value, the current path point can be selected as the sub-goal. This method helps to identify key points with significant turns and major route changes from the global path, thereby providing precise target nodes for subsequent local path planning. When the UAV encounters a local minimum, the sub-goal can guide the UAV to escape from the local minimum. The sub-goal provides new guidance rather than the endpoint during the planning process and helps the UAV avoid the local minimum trap.

Setting the initial orientation of the UAV
The correct initial orientation can ensure that the UAV remains stable during takeoff and flies smoothly towards the goal point. Assuming the current path point is $P_i(x_i, y_i)$ and the next goal point is $P_{i+1}(x_{i+1}, y_{i+1})$, the orientation θ_i of the current path point can be calculated using Eq. (27).

$$\theta_i = arctan(\frac{y_{i+1} - y_i}{x_{i+1} - x_i}) \tag{27}$$

RRT-DWA hybrid path planning algorithm process
The hybrid algorithm can efficiently and safely complete path planning tasks in complex dynamic environments, dynamically adapting to environmental changes, especially in environments with dynamic obstacles, ensuring the UAV's high-speed obstacle avoidance capability. The overall process is shown in Fig. 2.

Results and analysis

The experiments in this chapter were run on MATLAB 2023b, Intel Core i5-12400F, and 16GB RAM, focusing on verifying the path - planning performance of the RRT - DWA algorithm in complex environments and analyzing its performance in multi - task and dynamic obstacle scenarios.

Obstacle environment setup

The two-dimensional environment area designed in this chapter's simulation is a 100 m*100 m plane region, and the three-dimensional environment area is a 30 m*30 m*30 m cubic region, with the distribution of obstacles as shown in Fig. 3 and Fig. 4.

Verification and analysis

The DWA parameters are set as shown in Table 1.

In the two-dimensional environment, the start point is set at (10, 90), and the goal point is set at (85, 5). Path planning is performed using both the basic DWA algorithm and the improved DWA algorithm, with the results shown in Fig. 5. During the basic DWA planning process, the algorithm chooses to pass between two obstacles, as indicated by the red box in the figure. Although this path is relatively direct, the short safety distance between the two obstacles and the consecutive turns that do not conform to the UAV's flight manner may pose flight risks. In the improved DWA, by enhancing the obstacle avoidance strategy, the algorithm chooses to go around these two obstacles, thereby avoiding potential collision risks and also avoiding consecutive large-angle turns in the path, ultimately forming a safer and more feasible path.

The start point was reset to (0, 0) and the goal point to (85, 80). Path planning was conducted using both the basic DWA algorithm and the RRT-DWA hybrid path planning algorithm, with the results shown in Fig. 6. During the path planning process, the basic DWA algorithm encountered a local minimum at point (63, 54), causing the path planning to fail. The RRT-DWA hybrid path planning algorithm, by introducing global path guidance and defining sub-goal points, effectively avoided falling into local minima.

To check whether the algorithm can handle dynamic obstacles, the start points were set to (0, 100) and (0, 0), and the goal points were set to (85, 5) and (85, 85), respectively. In this experiment, no task allocation mechanism was used. Instead, the start point (0, 100) was mapped to the goal point (85, 5), and the start point (0, 0) was mapped to the goal point (85, 85), so that the paths of the two tasks would intersect during the planning process to verify the algorithm's real-time obstacle avoidance capability. The RRT-DWA hybrid path planning algorithm, by introducing a real-time obstacle avoidance strategy, successfully triggered the dynamic obstacle avoidance mechanism, reduced the flight speed, and adjusted the turning angle to avoid the collision threat, as shown in Fig. 7.

To test the target reachability of the algorithm, this paper conducted 100 path - planning experiments with random start and goal points. The success rate is 100% and average planning time is 3.43s . These results demonstrate the target reachability of the proposed algorithm in this environment.

The overall planning result of the RRT-DWA hybrid path planning algorithm is shown in Fig. 8.

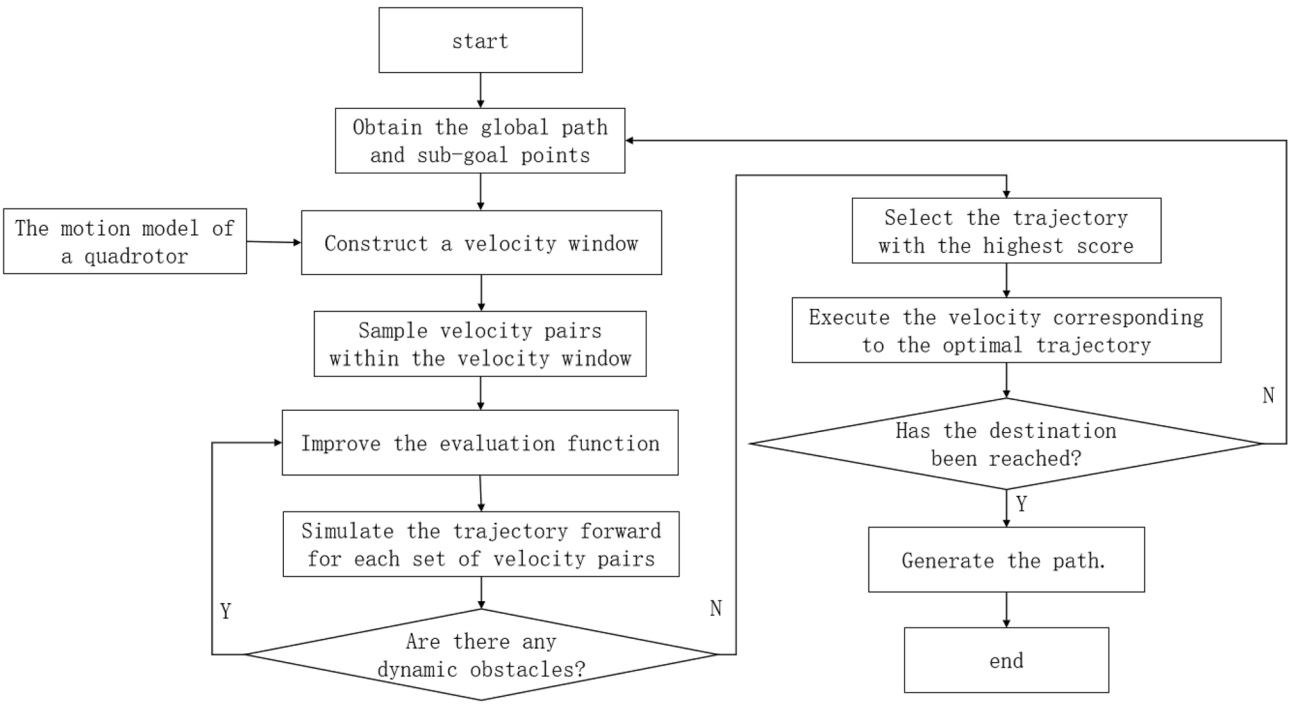


Fig. 2. Hybrid algorithm flow.

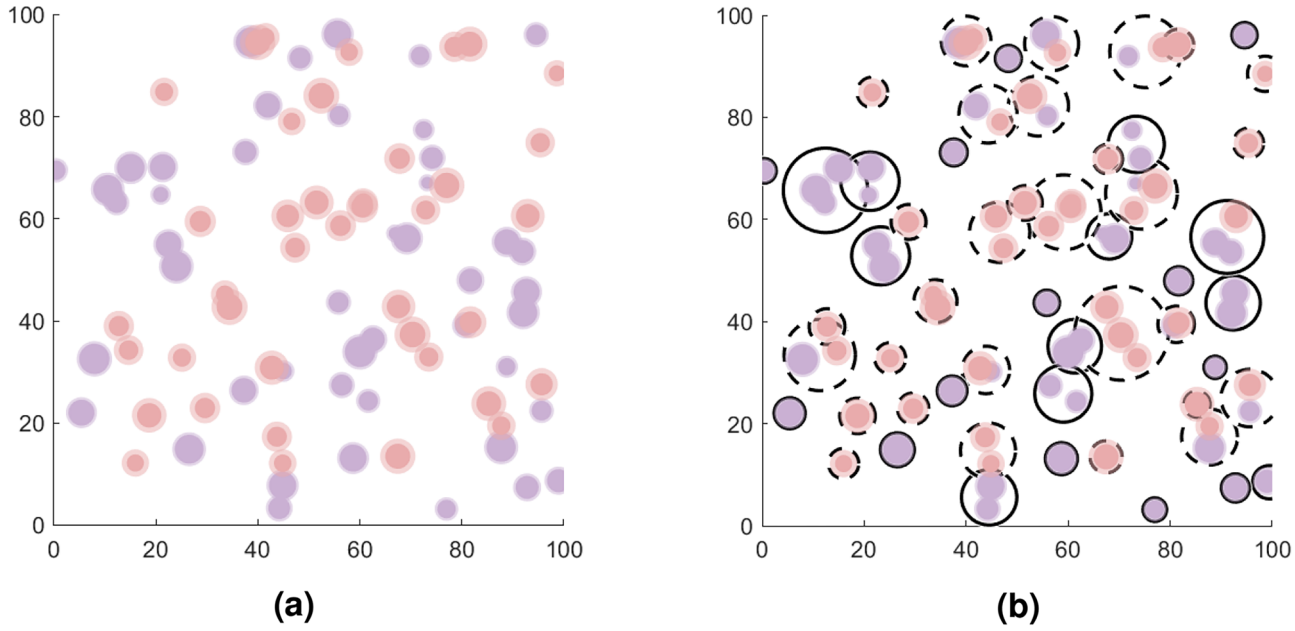


Fig. 3. Obstacle environment. (**a**) Two-dimensional Obstacle Distribution; (**b**) Two-dimensional Clustering Results.

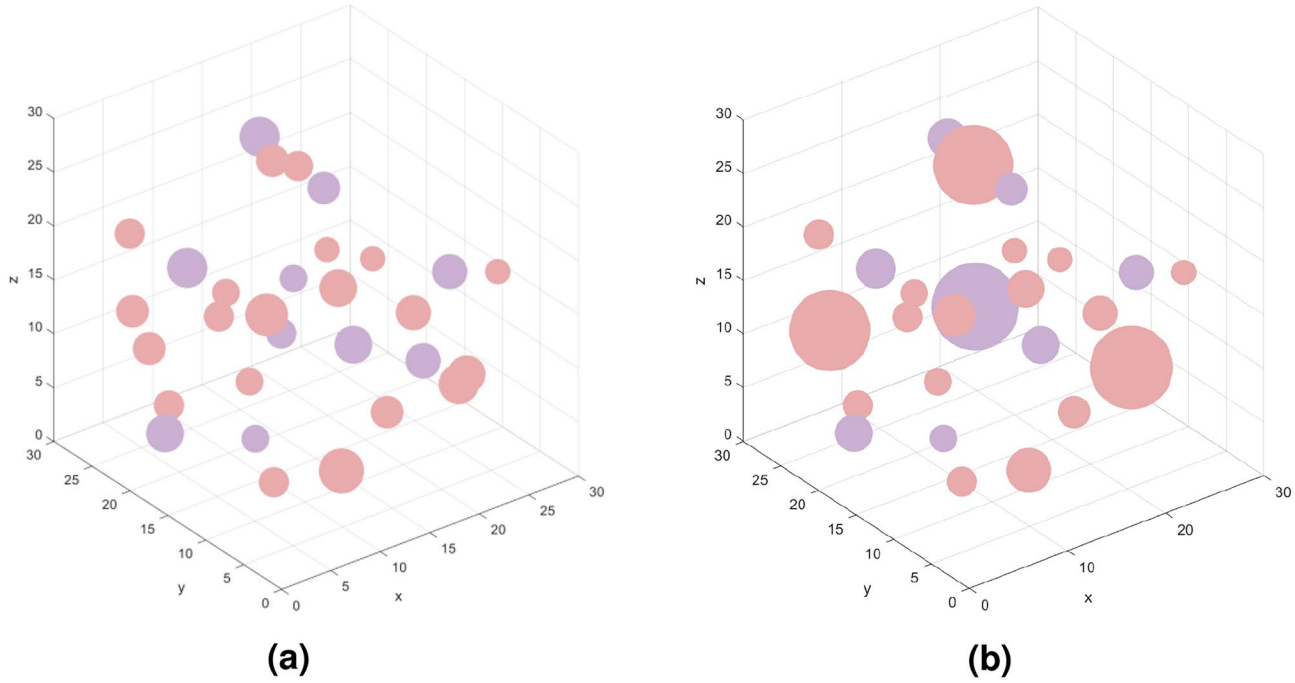


Fig. 4. Obstacle environment. (**a**) Three-dimensional Obstacle Distribution; (**b**) Three-dimensional Clustering Results.

Table 2 summarizes the path lengths and associated costs. In Task 1, the final path length of the improved RRT is 140.69 m with a cost of 399.30, while the final path length of RRT-DWA hybrid path planning algorithm is 133.75 m, representing an optimization of 4.93%. In Task 2, the final path length of the improved RRT is 150.95 m with a cost of 380.50, while the final path length of RRT-DWA hybrid path planning algorithm is 130.62 m, representing an optimization of 13.46%.

The start point was reset to (0, 100) with goal points at (85, 5) and (85, 85), respectively, as shown in Fig. 9. The yellow dynamic obstacle 1 started at (10, 100) and moved vertically downward at a constant speed of 2 m/s until it stopped at (10, 80). The red dynamic obstacle 2 moved back and forth between the positions (45, 50)

Parameter Name	Numerical
Maximum Linear Velocity	5.12 m/s
Maximum Linear Acceleration	1.05 m/s^2
Maximum Angular Velocity	$\pi/3$ rad/s
Maximum Angular Acceleration	$\pi/3$ rad/s^2
Forward Trajectory Prediction Time	2 s
Obstacle Threat Distance	1 m
Rigid Obstacle Threat Distance Weight	0.2 m
Flexible Obstacle Threat Distance Weight	0.5 m

Table 1. DWA basic parameter settings under single task.

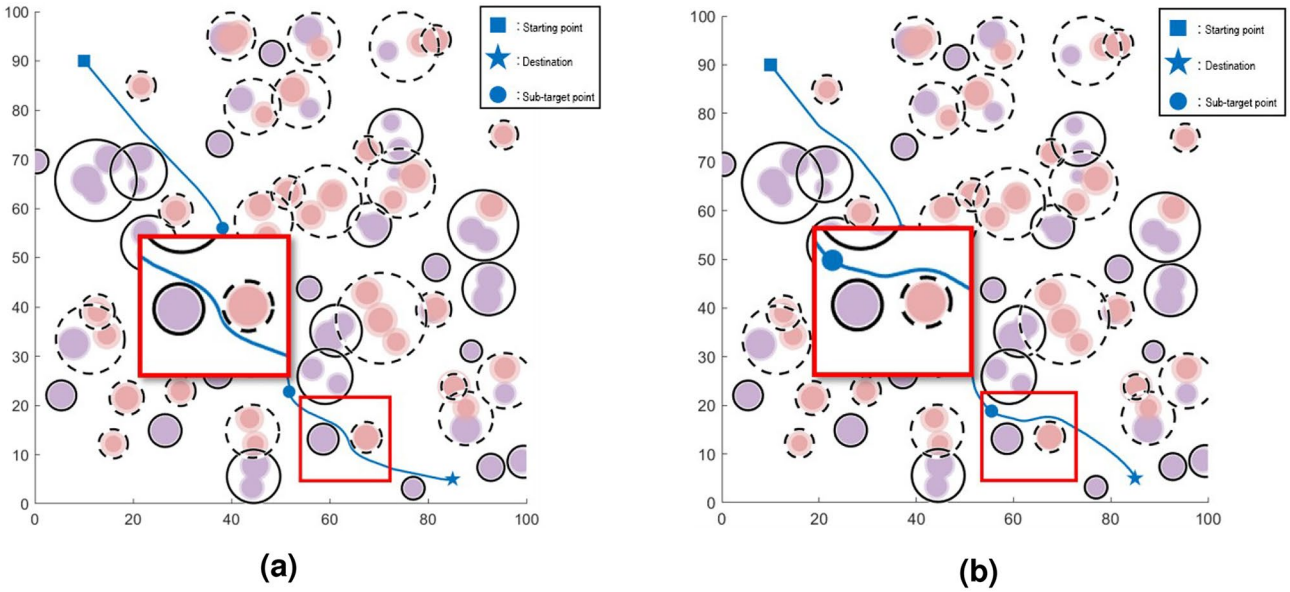


Fig. 5. 2D comparison of DWA results.(**a**) Basic DWA;(**b**) Improved DWA.

and (45, 40) at a speed of 2 m/s. During the simulation, the UAV was able to avoid the dynamic obstacles and overcome the threats to reach the target position.

The start points were reset to (5, 5), (10, 90), (55, 5), and (85, 85), and the goal points were set to (10, 50), (50, 45), (60, 50), and (95, 35), respectively. Path planning was performed using the RRT-DWA hybrid path planning algorithm, with the cost map and planning results shown in Fig. 10. The data are presented in Table 3. For Task 1, the final path length was 72.32 m, representing an optimization of 16.3% compared to the global planning length of 86.42 m using RRT. For Task 2, the final path length was 74.89 m, representing an optimization of 12.3% compared to the global planning length of 85.41 m using RRT. For Task 3, the final path length was 51.93 m, representing an optimization of 6.02% compared to the global planning length of 55.26 m using RRT. For Task 4, the final path length was 55.93 m, representing an optimization of 7.15% compared to the global planning length of 60.24 m using RRT.

To validate the feasibility and superiority of the algorithm in a larger space, this paper conducted comparative experiments with the RRT* algorithm in a 100*100 m space, and the results are shown in Fig. 11 and Table 4. RRT-DWA achieved the shortest average path length (57.50 meters), validating the effectiveness of locally optimizing the global path through the dynamically weighted evaluation function and the key sub-goal guidance mechanism. In terms of path smoothness, although its average curvature (0.068) is slightly higher, its maximum curvature (0.73) is significantly lower than that of RRT* (1.092), reflecting the positive role of the improved velocity smoothness evaluation function in DWA, which makes the generated path more trackable for UAVs. Although its average planning time (3.52 seconds) increased due to the introduction of local replanning and dynamic prediction, the overall improvement in path length and smoothness demonstrates the superiority and practicality of the algorithm in complex dynamic environments.

To systematically evaluate the influence of the dynamic weighting parameters in the enhanced DWA algorithm on planning performance, we conducted a sensitivity analysis using the same obstacle environment as in Fig. 11, with start and goal points set to (0,100) and (100,0), respectively. The experiments focused on varying the weights assigned to goal distance, obstacle avoidance, and velocity smoothness, with results summarized in Fig.

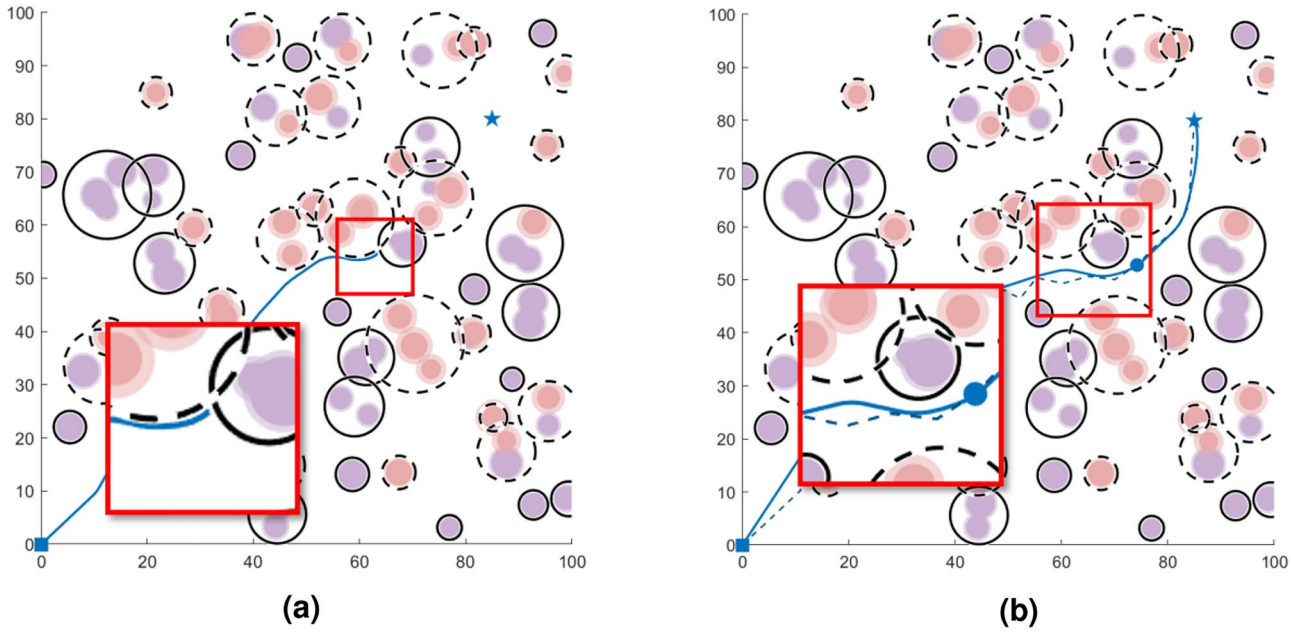


Fig. 6. Local minimum problem. (**a**) Basic DWA; (**b**) RRT-DWA.

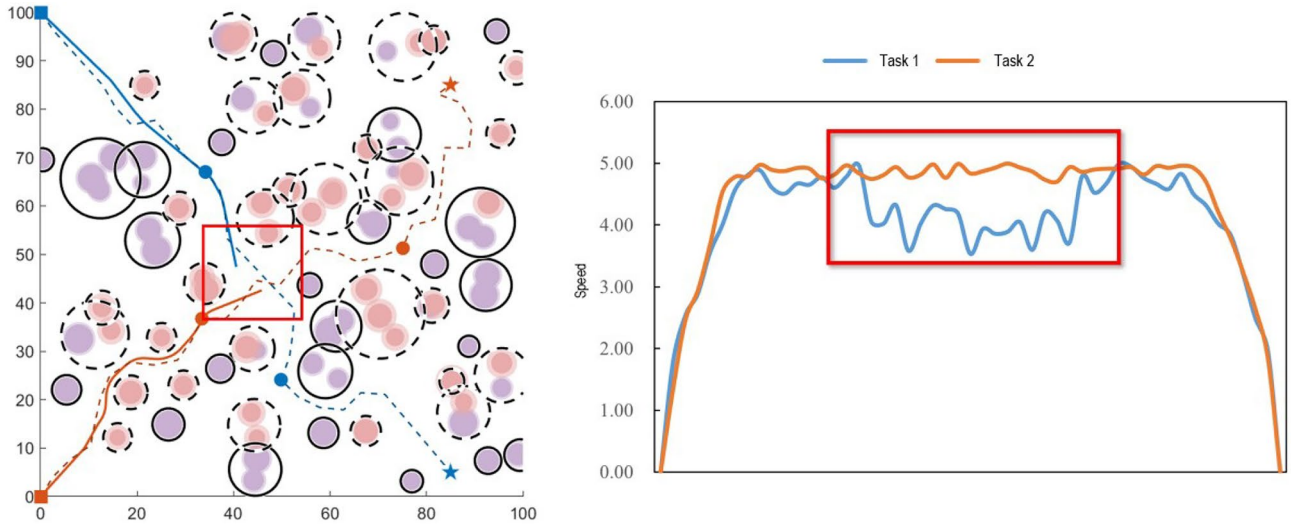


Fig. 7. Flying object collision threat avoidance.

12. Experimental findings demonstrate that these weighting parameters significantly affect path length, success rate, computation time, and overall evaluation score across different task scenarios. Specifically, increasing the goal-distance weight encourages more direct movement toward the target, thereby reducing overall path length. However, an excessively high value can increase collision risk in confined spaces, lowering the obstacle-avoidance success rate. Elevating the obstacle-avoidance weight substantially improves the system's responsiveness to both static and dynamic obstacles, raising success rates notably in densely cluttered environments. Nevertheless, an overly high setting may lead to circuitous paths and greater computational load. The velocity-smoothness weight regulates path quality and flight stability; while its variation has a comparatively modest impact on success rate, setting it too high can induce trajectory oscillations and a decline in the overall evaluation score.

Verification was conducted in the three-dimensional obstacle environment set in Section 3.1. The start point was set to (0, 0, 0) and the goal point to (30, 25, 15), with the planning results shown in Fig. 13. The length and cost of the path in the figure are shown in Table 5. For Task 1, the final path length was 30.84 m, representing an optimization of 21.6% compared to the global planning length of 39.31 m using RRT. For Task 2, the final path length was 56.14 m, representing an optimization of 22.6% compared to the global planning length of 72.49 m using RRT. For Task 3, the final path length was 31.75 m, representing an optimization of 11.7% compared to the global planning length of 35.96 m using RRT.

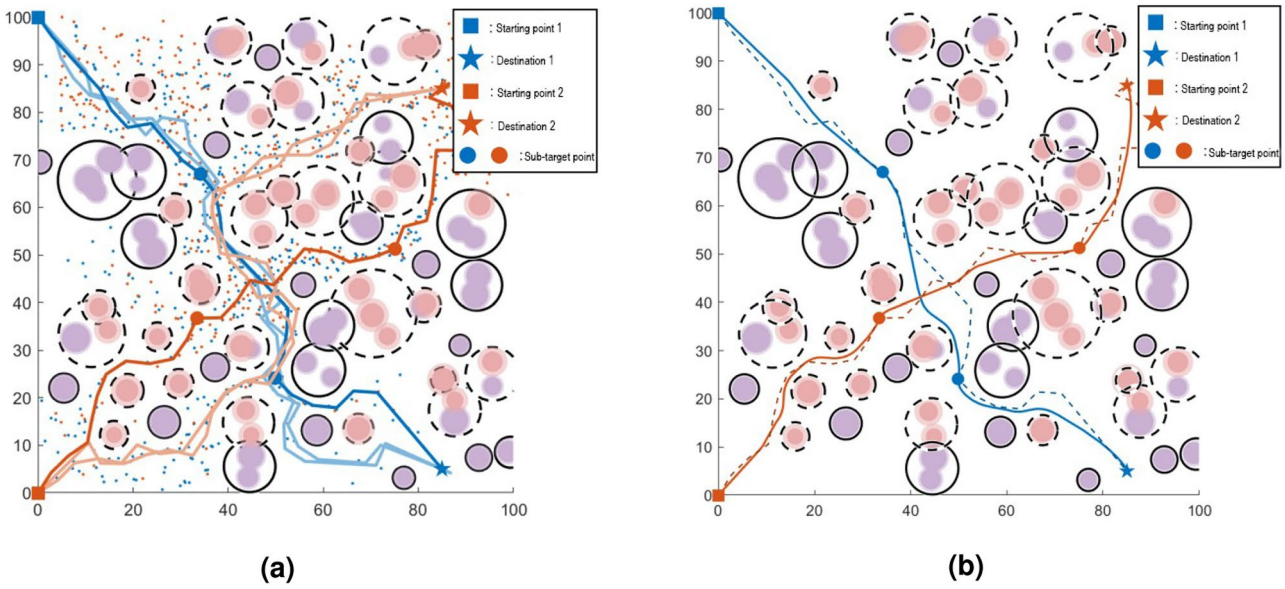


Fig. 8. 2D overall planning results.(**a**) Global Planning;(**b**) RRT-DWA.

Task	Length of RRT(m)	Cost	Final Path of RRT-DWA(m)
Task 1 Path 1	**140.69**	**399.30**	
Task 1 Path 2	160.89	475.81	133.75
Task 1 Path 3	156.79	444.82	
Task 2 Path 1	166.29	422.66	
Task 2 Path 2	**150.95**	**380.50**	130.62
Task 2 Path 3	160.49	386.48	

Table 2. 2D RRT-DWA simulation result data.

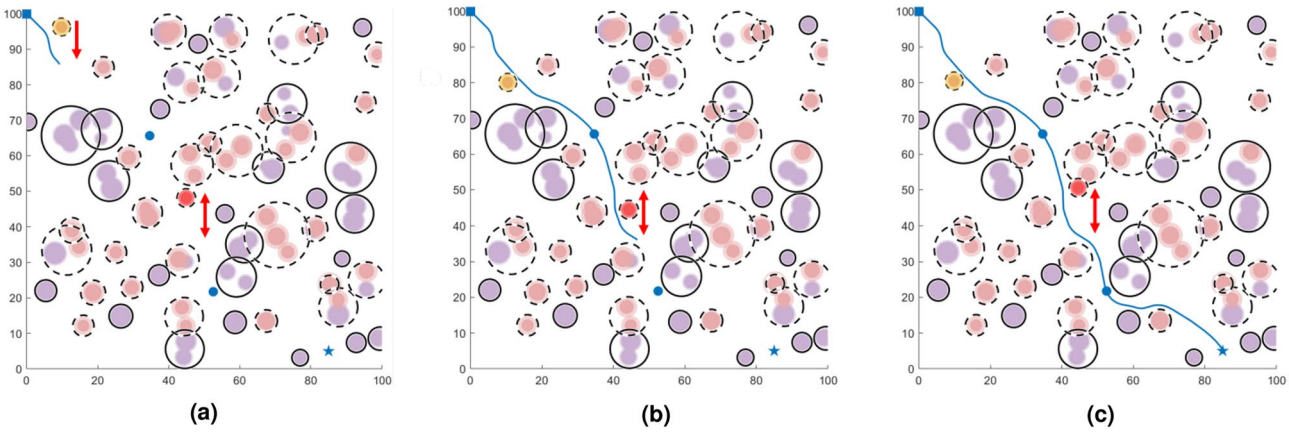


Fig. 9. Dynamic obstacle threat avoidance.

To validate the efficacy of the proposed algorithm in a physics-based simulation environment, experimental tests were conducted on TheConstruct platform utilizing ROS and Gazebo. A Parrot AR.Drone model was employed within a 5 m*5 m arena populated with 15 randomly distributed cylindrical obstacles.

The algorithm was implemented as a set of modular ROS nodes: the global RRT planner operated as a service node to generate an initial reference path at mission start, while the enhanced DWA local planner subscribed in real time to $/drone/gt_pose$ for odometry data and to $/gazebo/model_states$ for LiDAR-based environmental perception. Velocity commands were subsequently published to the $/cmd_vel$ topic to control the drone's motion.

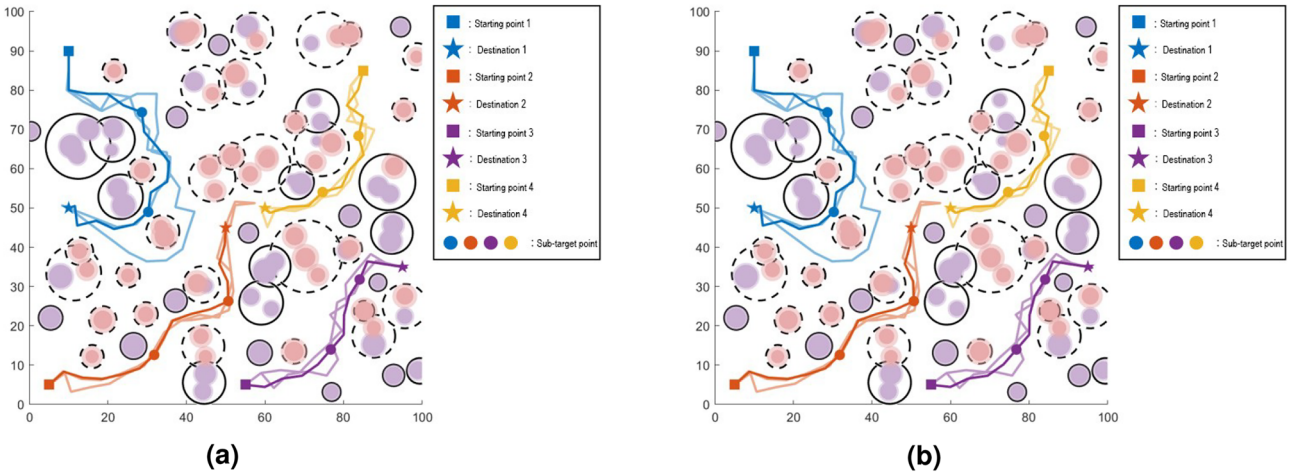


Fig. 10. 2D multi-tasking planning results.(**a**) Global Planning;(**b**) RRT-DWA.

Task	Length of RRT(m)	Cost	Final Path of RRT-DWA(m)
Task 1 Path 1	87.51	183.21	
Task 1 Path 2	**86.42**	**180.47**	72.32
Task 1 Path 3	100.23	203.15	
Task 2 Path 1	**85.41**	**168.41**	
Task 2 Path 2	93.21	180.32	74.89
Task 2 Path 3	91.54	177.24	
Task 3 Path 1	**55.26**	**105.65**	
Task 3 Path 2	55.51	109.05	51.93
Task 3 Path 3	60.81	114.65	
Task 4 Path 1	66.13	300.13	
Task 4 Path 2	61.16	289.42	55.93
Task 4 Path 3	**60.24**	**287.85**	

Table 3. 2D RRT-DWA simulation result data.

Experimental results (Fig. 14) indicate that the drone successfully navigated from the start point (0,0) to the goal (5,5). Upon encountering unforeseen dynamic obstacles, the algorithm activated a real-time collision avoidance mechanism, producing smooth local trajectory adjustments. Across 10 independent trials, the mission success rate reached 100%, confirming the robustness and practical applicability of the proposed approach in a physics-based simulated environment.

Conclusion and future work

This paper proposes a hybrid RRT-DWA path planning framework for UAVs that advances beyond existing implementations through several key innovations. Unlike the approach of Li et al.[25], which mainly uses the global path as a directional constraint, our method establishes a dynamic feedback loop between the global and local planners. The introduced sub-goal selection mechanism not only avoids local minima but also continuously refines path quality based on real-time local observations.

Compared to learning-enhanced DWA approaches[31,34], our dynamic weighting strategy provides a computationally efficient alternative that doesn't require offline training or extensive datasets. While learning-based methods excel in complex environments, they often face deployment challenges in real-world UAV applications due to their computational demands and generalization concerns. Our method maintains the interpretability and reliability of traditional DWA while significantly enhancing its adaptability through principled mathematical formulations rather than black-box optimization.

Simulations demonstrate the framework's superiority over conventional methods in both path quality and dynamic obstacle avoidance.

Limitations and future work

Despite its performance, the framework has limitations that open avenues for future research. Primarily, the increased computational load from the hybrid structure and dynamic prediction may challenge the real-time capabilities of resource-constrained UAV platforms. Furthermore, while validated in 3D space, the planner's core logic operates effectively in 2.5D, and its extension to fully complex 3D navigation (e.g., with overhangs) requires

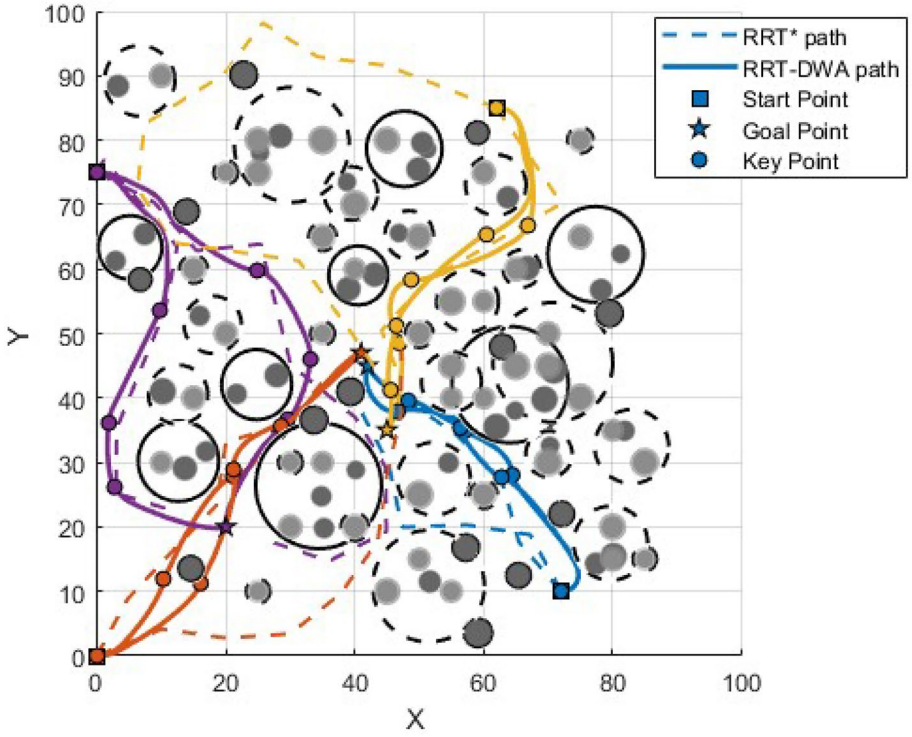


Fig. 11. Comparison experiments between RRT-DWA and RRT*.

Algorithm	Average Planning Time(s)	Average Path Length(m)	Average (Maximum) Path Curvature
RRT	0.93	80.88	0.052(0.83)
RRT*	2.66	64.09	0.056(1.092)
RRT-DWA	3.52	57.50	0.068(0.73)

Table 4. The results of the comparison experiments between RRT-DWA and RRT*.

integration with a 3D local map. Finally, the reliance on precise environmental data in simulation necessitates future work on robustness to real-world sensor noise and uncertainty.

Accordingly, our immediate future work will focus on: (1) optimizing the algorithm's computational efficiency for embedded deployment; (2) extending the local planner to leverage full 3D environmental information;(3) validating the framework on a physical UAV platform under real-world perception challenges.

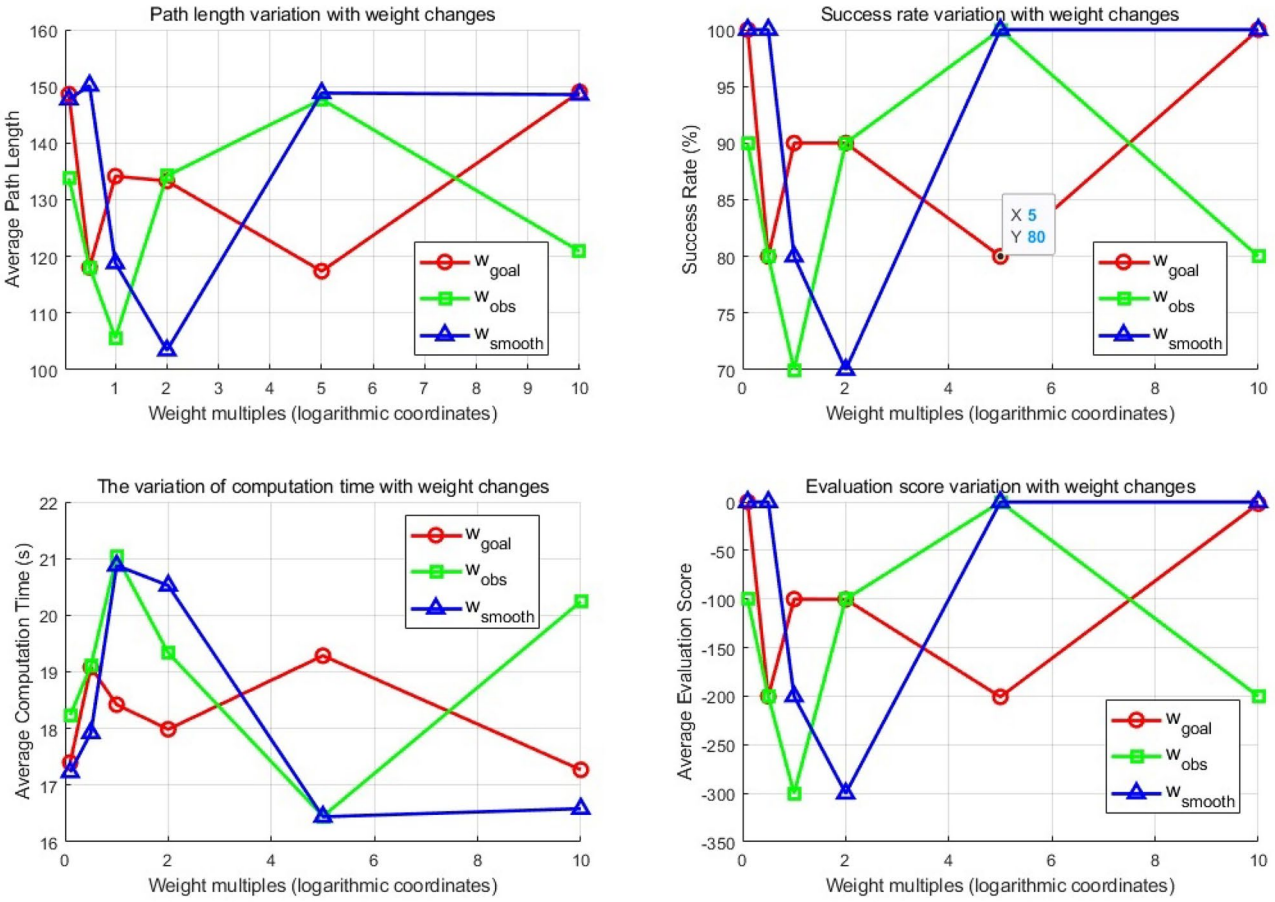


Fig. 12. Sensitivity analysis of algorithm weight parameters.

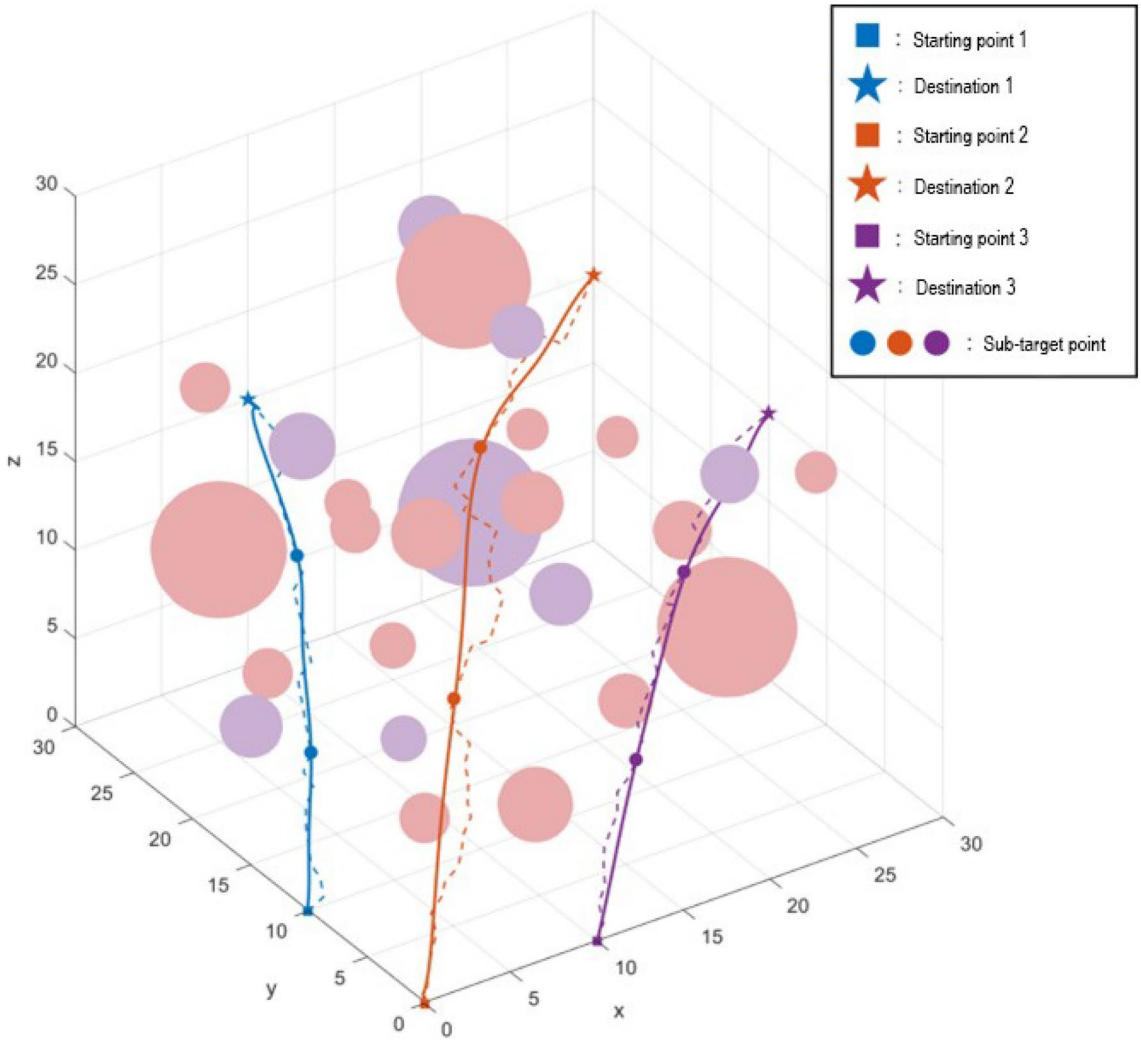


Fig. 13. 3D RRT-DWA planning results. (All axes are in meters (m)).

Task	Length of RRT(m)	Cost	Final Path of RRT-DWA(m)
Task 1	39.31 ±2.31	315.18 ±22.1	30.84 ±1.66
Task 2	72.49 ±4.26	451.54 ±32.5	56.14 ±2.14
Task 3	35.96 ±2.64	225.43 ±16.4	31.75 ±1.85

Table 5. 3D RRT-DWA simulation result data.

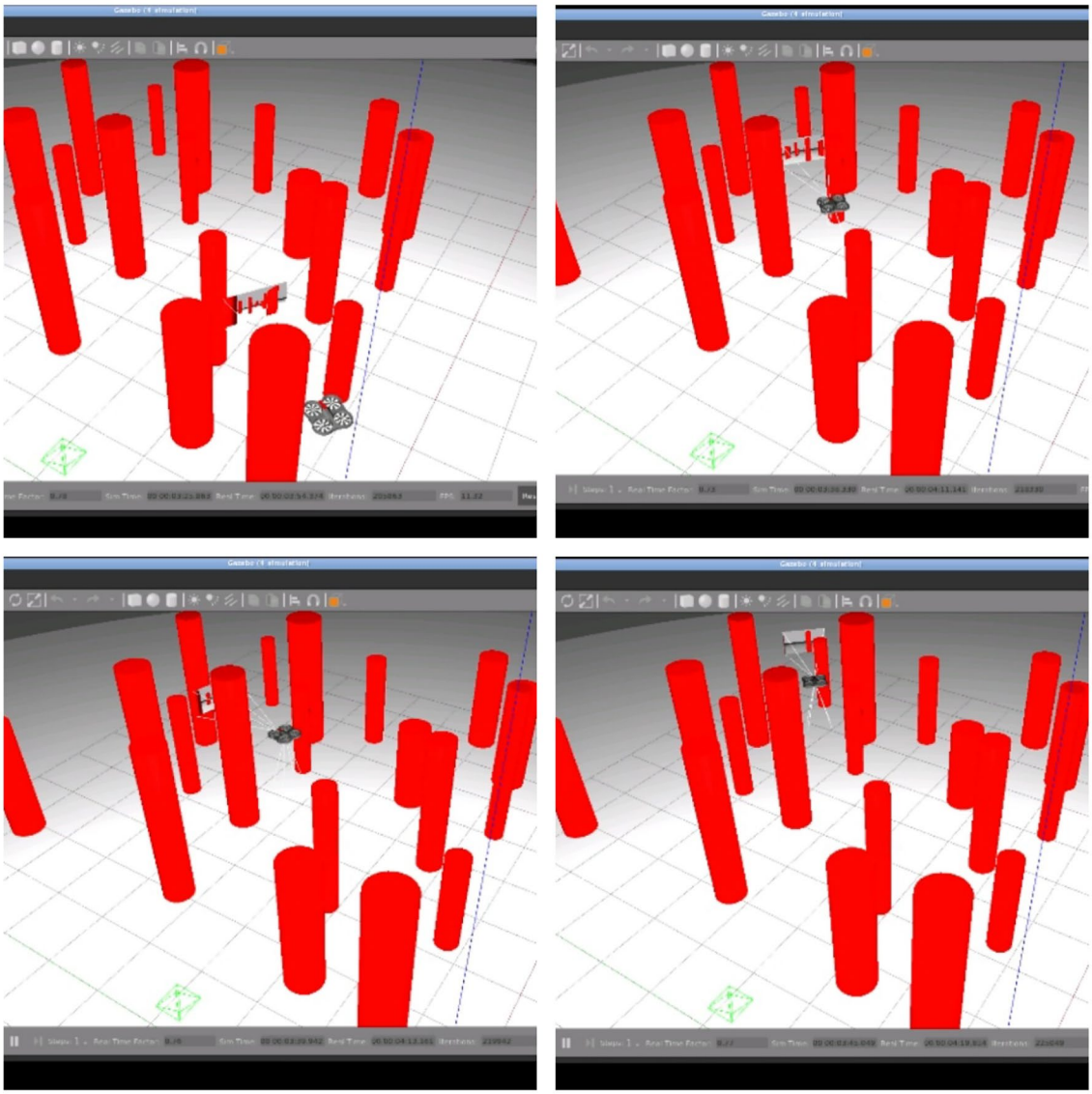


Fig. 14. ROS-Gazebo simulation experiments.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

Conceptualization, X.M. and Q.H.; methodology, Q.H., X.M.; software, J.L.; validation, X.M., H.L.; formal analysis, Q.Y.; writing—original draft preparation, X.M.; writing—review and editing, Q.H., Y.Y.; visualization, X.M.; supervision, Q.H.,Q.Y; project administration, Q.H.; funding acquisition, Q.H.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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